



LIFTING OPERATIONS

METHOD STATEMENT / LIFT PLAN & RISK ASSESSMENT

DETAILS FOR THE PROVISION OF A CONTRACT LIFT OPERATION IN COMPLIANCE WITH
BS7121 – PARTS 1 & 3, LOLER AND POWER REGULATIONS

Client	<i>Weatherproofing</i>
Project	<i>Lifting roofing materials to roof level</i>
Site Address	<i>Iron Mountain, 9 White Hart Avenue, London, SE28 0FD</i>

Date Valid From/To	<i>31/07/2023</i>	<i>31/07/2023</i>	Time on site	<i>08:00</i>
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M/S No.	<i>SG2307213</i>	Rev No.	<i>01</i>
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Client / site contact	<i>Ben Tough</i>
Contact Number	<i>07540 083359</i>

Details for Mobile Crane Operator / Lifting supervisor

Crane Details	<i>Tadano Faun AFT70G-4L</i>
Alternative Crane	<i>tba</i>
Loading For City Lifting use only	<i>No Loading</i>
Access - Egress	<i>Crane to enter site off White Hart Avenue</i>
Other	<i>No loads heavier than 200kg to be landed on the roof</i>

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Table of Contents

Introduction

- 1. The Contract lift.**
- 2. General Conditions**
- 3. Ground Conditions**
- 4. Personal Protective Equipment**
- 5. Operation Outline**
- 6. Weather Conditions**
- 7. Site Details**
- 8. Crane Details**
- 9. Load Details**
- 10. Crane specification chart**
- 11. Variation to Method Statement**
- 12. Placement of additional mats**
- 13. Crane set up procedure**
- 14. Lifting Procedure**
- 15. De-Rig Procedure**
- 16. Emergency Procedure**
- 17. Risk Assessment 17.1 - 17.12**
- 18. Toolbox talk attendance sheet**
- 19. Lifting Plans and Supporting Documentation**

Introduction

This Method Statement has been prepared and based on the information supplied by Weatherproofing together with the Site Inspection, Risk Assessment Report and Sketches and / or Drawings completed by the City Lifting Ltd Appointed Person.

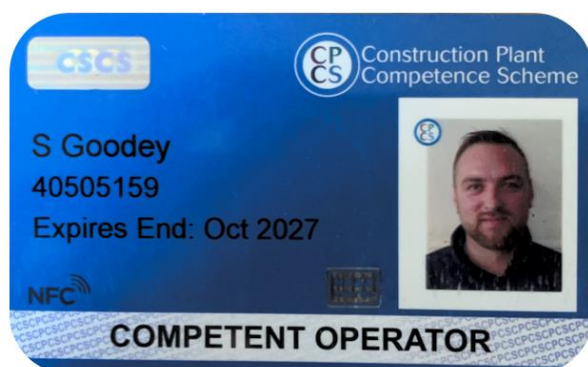
All work will be subject to CPA Conditions for Contract Lifting and will be performed in accordance with:

- The Management of Health and Safety at Work Regulations 1999 Reg. 3.
- Health & Safety at Work Act 1974
- BS 7121 Safe Use of Cranes and Lifting Operations Part 1 2016 and Part 3 2012
- Lifting Operations and Lifting Equipment Regulations 1998.
- The Provision and Use of Work Equipment Regulations 1998 Reg. 3 to 11.
- The Work at Height Regulations 2005
- Coshh (supplied with each individual crane on site)

Appointed Person's Acceptance of Responsibilities.

I confirm that the lifts in this Method Statement have been planned and will be carried out in accordance with current legislation and British Standard 7121 and that I accept responsibility for the preparation of this Method Statement and Risk Assessment. I accept that it is my responsibility as the Appointed Person to ensure that the Lift Supervisor is adequately briefed on the contents in this Method Statement.

Appointed Person	<i>Steve Goodey</i>
CPCS No.	<i>40505159</i>
Mobile	<i>07484 061183</i>



COMPETENT OPERATOR

CATEGORIES

A40A: Slinger / Signaller - All Types - All Duties
A60C: Mobile Crane - All Duties
A61: Appointed Person - Lifting Operations
A62: Crane/Lifting Operations Supervisor

This card is issued in accordance with the terms laid out in the CPCS Scheme Booklet.
CPCS is part of the NOCN Job Cards

Date created	<i>21/07/2023</i>
Date Amended	Click or tap to enter a date.
Signed	<i>SR Goodey</i>

1. The Contract Lift

The Contract Lift will be carried out in accordance with CPA Standard Terms and Conditions for Contract Lift involving the Lifting and Movement of Goods for a Crane Operation.

1.1 SITE INSPECTION

The Appointed Person shall carry out a detailed site inspection to ascertain the full requirements of the Client determining the required crane capacity, including an assessment as far as possible / practicable for any other site issues such as access, ground conditions (visual inspection or site survey results), overhead obstructions, local authority interaction etc.

1.2 THE APPOINTED PERSON

The Appointed Person who has prepared this Method Statement carries full responsibility for the safe completion of all works carried out during the operation. It is the responsibility of the Appointed Person to ensure that the Lift Supervisor is adequately briefed on the contents of this Method Statement.

1.3 THE LIFT SUPERVISOR

Prior to the commencement of the works, the Lift Supervisor must ensure that all site personnel are adequately briefed on the contents of this Method Statement. This briefing shall take the form of an adequate "task briefing". The Lift Supervisor must liaise with the Appointed Person should site circumstances require material change to the methods to be employed during the operation.

1.4 THE CRANE OPERATOR

All City Lifting operators or any operators working on behalf of City Lifting shall be fully trained and competent on the machines they operate and will have undergone a full company induction.

1.5 PERSONNEL

All City Lifting personnel or any personnel working on behalf of City Lifting, will be fully trained and hold the relevant certification for the role they undertake on the day of operation. Any non-City Lifting personnel or personnel supplied by Weatherproofing who will be involved in the lifting operation and taking on a specified role, will be required to produce the required certification to the Appointed Person or Lift Supervisor prior to the commencement of the operation. Failure to produce the required certification will result in the suspension and possible cancellation of the lifting operation.

1.6 CERTIFICATION

Copies of statutory test certification and inspection reports for all City Lifting supplied equipment and personnel will be made available for inspection to Weatherproofing. The certification will be supplied at the site induction phase of the day/scenario prior to the lifting operation commencing.

2. General Conditions

- 2.1** Weatherproofing will be required to provide details of all items to be lifted, including weights, dimensions and specific lifting tackle requirements. Weatherproofing is responsible for the structural integrity of all loads to be lifted including any specified lifting points or slinging recommendations.
- 2.2** Weatherproofing must ensure safe access to and egress from site for all City Lifting personnel and vehicles, and also ensure that the working area is clear of all obstructions to allow the safe assembly and working of the crane. Weatherproofing will be responsible for supplying suitable floodlighting (C.P.A terms & conditions) and must carry out any enabling works required prior to crane arrival. The Lift Supervisor must have adequate parking on site for the duration of the job to enable the required lifting tackle to be supplied to the lifting operation.
- 2.3** Weatherproofing must ensure that the area where the crane is to stand / operate is firm and level and that it is capable of withstanding the axle and outrigger loadings that will be imposed upon it.
- 2.4** Weatherproofing must ensure that any Traffic Management not supplied by City Lifting is in place prior to the positioning of the crane
- 2.5** Any personnel not directly concerned with the lift must be kept out of the crane assembly and working area. The working area will be cordoned off with temporary fencing or otherwise secured to prevent non-lift associated personnel or the general public from entering the lifting area.
- 2.6** The Lift Supervisor shall make themselves known to all relevant personnel on site, and conduct a task briefing prior to the lifting operation commencing. Weatherproofing should arrange for a site-specific induction for City Lifting personnel prior to the commencement of any work. The Client / Site Risk Assessment, which identifies any site hazards should be made available and brought to the attention of all City Lifting personnel.
- 2.7** All cranes to be left unattended for any period shall be left in safe condition with no load on the crane hook. Cranes shall be shut down and isolated as per manufacturer's instructions, secured and locked. Under no circumstances shall the crane be moved for lifting purposes if not under the instruction of the Lift Supervisor.
- 2.8** In the event that specified cranes or lifting tackle cannot be supplied on the day of operation due to operational issues, breakdown or any unforeseen circumstances, City Lifting reserves the right to supply cranes or equipment that is of a suitable alternative to that detailed in this Method Statement. In the event of this situation arising, the Appointed Person and Weatherproofing will be notified. The AP / Lift Supervisor will ensure that any changes do not infringe on the factors of safety within this Method Statement.

3. Ground Conditions

- 3.1** City Lifting will endeavour to advise the client regarding ground conditions and supply suitable mats where required. However, City Lifting will take no responsibility whatsoever for the ground conditions. As Weatherproofing are entirely responsible for ground conditions and should ensure that the ground where the crane is to stand is capable of withstanding the outrigger and axle loads stated in this method statement. Weatherproofing are also required to advise City Lifting of any known underground services, voids or vaults as stated in BS7121. Safe Use of Cranes.

4. Personal Protective Equipment

- 4.1** The appropriate PPE, as determined by the Risk Assessment and the Client's specific requirements, will be worn: Hard Hats EN50365, Orange Hi Vis Vests EN20471, Steel Toe lace up Boots EN20345, Gloves EN388, Safety Glasses EN166. Disinfectant wipes, surgical gloves

5. Operation Outline

5.1

Lift Classification			
Basic Lift	Intermediate Lift		Complex Lift
	Yes		

Full Contract Lift	Yes	Supervisor Only Contract Lift	
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If a "Supervisor Only Contract Lift" is the chosen option, it is imperative that any personnel supplied by the Client, be fully trained, competent and experienced in lifting operations and hold a current, valid Certification, which must be available for our Appointed Person or Lift Supervisor to inspect prior to commencement of the lifting operation. Any personnel supplied by the Client must work under the instruction of our Appointed Person or Lift Supervisor. Failure to comply with the above will result in the suspension of lifting until the afore- mentioned is adhered to. City Lifting reserves the right to suspend the lifting operation if any personnel supplied by Weatherproofing are deemed unsuitable.

5.2 PERSONNEL

City Lifting Ltd will provide the following numbers of personnel for this Contract lift.

Crane Operator	1	Lifting Supervisor	1	Slinger / signaller	1
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5.3 COMMUNICATION

Communication between the lifting crew and the crane operator will be maintained by the means of:

Two-way radios supplied by City Lifting	Yes	Two-way radios supplied by Client	No	Hand Signals	Yes
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The radios are to be tested for clarity of communication and correct channel selection before the lifting operation starts. Spare batteries are to be fully charged and the Crane supervisor/crane co-ordinator & slinger signaller must carry a spare battery with them at all times.

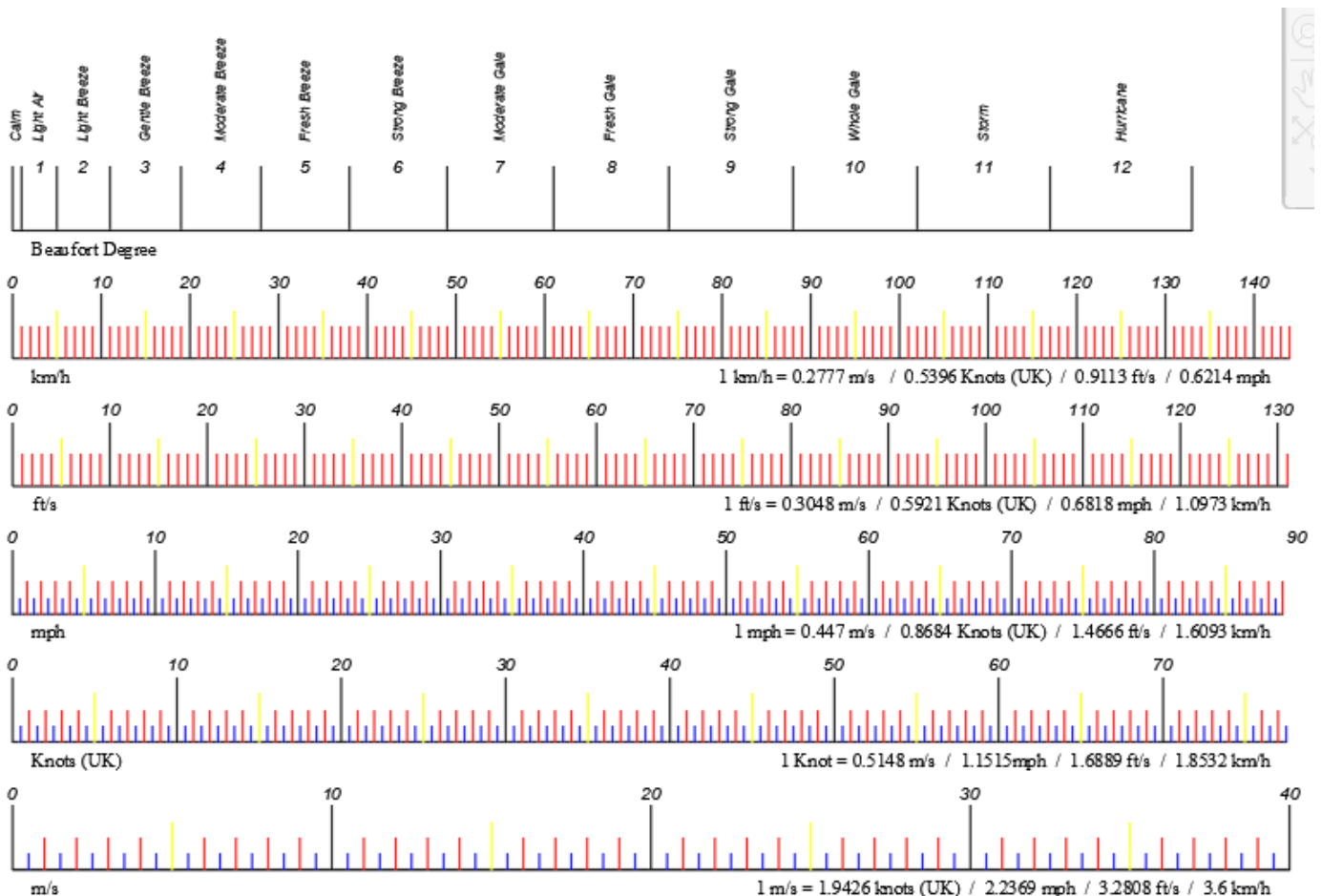
Radio communication is to be constant during instruction to the crane operator, if communication is lost the crane operator must stop immediately until communication continues.

6. Weather Conditions

In the event of extreme weather conditions City Lifting reserves the right to suspend the lifting operation until such time as it is safe to proceed. Although the safe workable wind speed varies from crane to crane, as a guide, the maximum wind speed at which lifting operations can take place is 9.8 metres per second, (22mph). In the event that man riding duties are required the maximum workable wind speed reduces to 7 metres per second (16 mph). (See section 6.1 for the cranes maximum wind speeds)

Please note that local conditions and the safety of personnel may dictate that the Lift Supervisor and Crane Operator have to suspend lifting operations at a lower wind speed than specified in section 6.1 of this method statement. In addition, lifting operations must stop during storms or when there is a risk of lightning strikes.

Maximum Wind Speeds	
Mobile Telescopic	9.8 metres per second
Man riding duties	7 metres per second



Windspeed Chart

7. Site Details

7.1 ENABLING WORKS

Any enabling works discussed and agreed to at the site inspection, should be carried out prior to the arrival of the crane. Enabling works required are as follows:

Area where crane is setting up to be clear f all vehicles and materials

7.2 ACCESS AND EGRESS

Weatherproofing should ensure that the crane and all associated vehicles have safe and unimpeded access and egress to and from site.

As page 1

7.3 GROUND CONDITIONS

Any underground services voids or vaults information supplied to the Appointed Person must be highlighted to the Lifting Supervisor. Please see section 8.2 for maximum outrigger loadings

Loadings are stated in section 8.2 and must be confirmed by WP & the Main contractor that the ground is capable of taking these loadings.

7.4 OVERHEAD OBSTRUCTIONS AND SLEWING RESTRICTIONS

Details of any overhead or slewing restrictions the lifting team should be aware of.

Crane team to be made to be aware of all overhead obstructions when slewing, crane team to set perimeter limiter to zone out possible interface between the loads lifted and the rest of the public

Loads are never to be taken out of site of the driver at any time. If this does happen the Slinger signaller must have eyes on the load and signal the driver in till he is in a position to see the load again.

7.5 TRAFFIC AND PEDESTRIAN MANAGEMENT

Details of any restrictions and local authorities concerned.

Traffic management required		Pedestrian management required	
No		No	
Supplied by Weatherproofing		Supplied by Weatherproofing	
Supplied by City Lifting		Supplied by City Lifting	

7.6 BRIEF SCOPE OF WORKS

Crane to set up in the specified location/s as per the supplied lifting plans- I have walked the crane position with the site supervisor and this area has been checked visually for the location of the mats and the outrigger loadings are detailed within the plans.

Once permission to proceed is gained, we will then bring the crane into the set-up area and close the area ready to set up and lift the various items to roof level.

All Transport will be supplied with edge protection by the clients transport company, in the event that edge protection isn't available the crane supervisor will place an inertia reel on the cranes hook and the slinger will wear a harness and attach their lanyard to the inertia reel before climbing onto the lorry. The slinger must stay clipped on at all times whilst working at height and ladders must always be footed when in use.

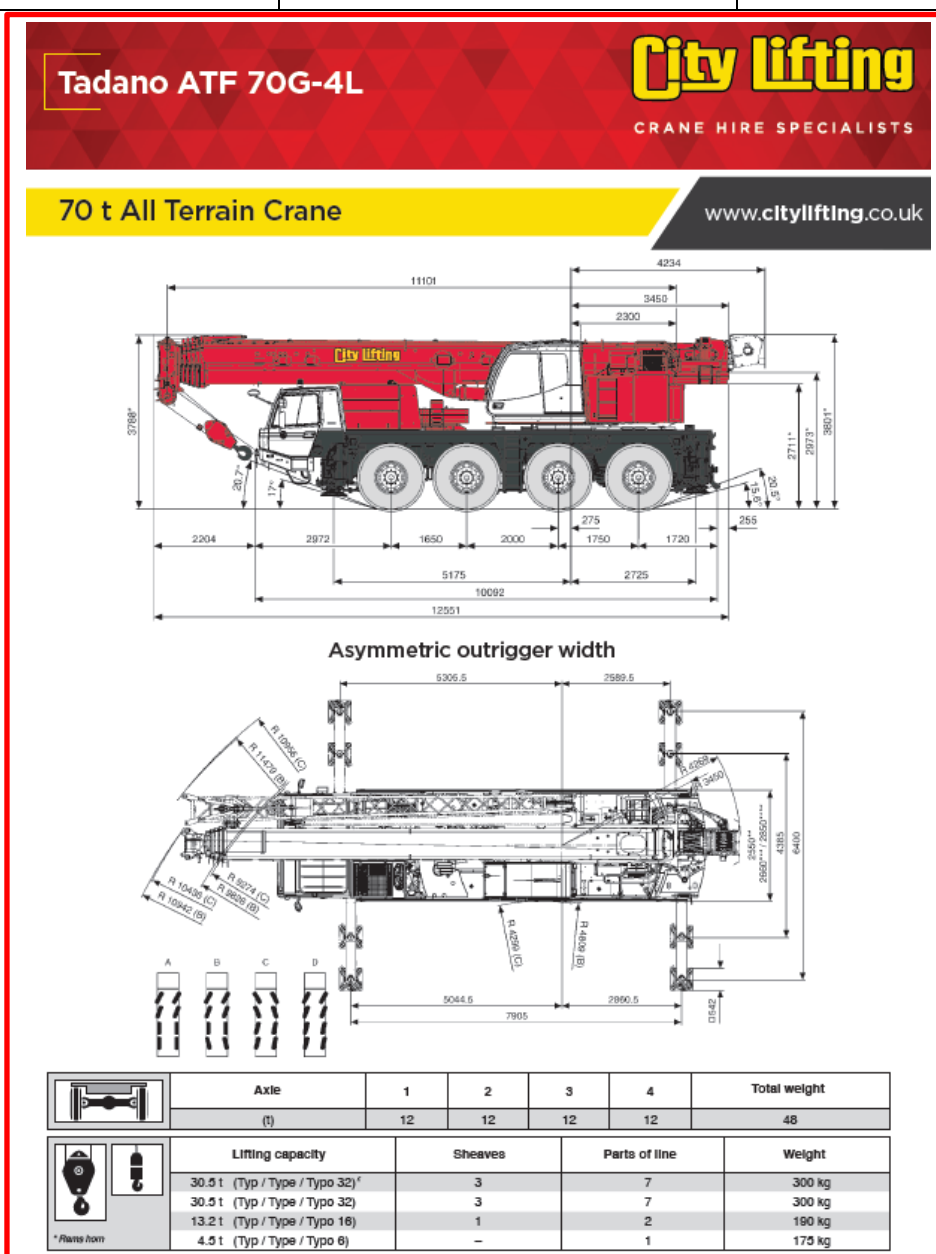
Once the job is completed and client is finished with the crane the crane operator will follow the de-rig procedure and leave site.

8. Crane Details

In exceptional circumstances, City Lifting Ltd reserves the right to change / rehire the proposed crane and if this becomes necessary our client will receive an updated Method Statement. It is our intention to use the following crane:

8.1 CRANE SPECIFICATION

	Main Crane	Choose an item.
Type	Tadano Faun AFT70G-4L	
Capacity	70te	
Outrigger Dimensions	7.90m x 6.40m	
Boom length	52.1m	
Fly jib length (x°)	n/a	
Counterweight	16.5te	
Hook block capacity	13.2te	
Hook block weight	190kg	



8.2 MAT SIZE, AXLE AND OUTRIGGER LOADS SEE ATTACHED LOADING SHEET

Max outrigger Load	22te	
Mat size - Type	1.2m Diameter	
Mat area m ²	1.13m ²	
Loadings per m ²	19.45te/m ²	
Gross vehicle weight	48,000kg	
Axle weights 1,2,3 & 4	12,000kg	



City Lifting



OUTRIGGER REACTION

ATF 70G-4 EU Stage IV (52,1m) (Standard)
Boom without Jib

CONFIGURATION

Jib - Configuration [m]	0	Slewing angle	Maximum ground bearing load in service in working area	Number of ropes	1
Jib - Adjustment angle [°]	0	Hook block [t Lifting load]	Included in load	Usable rope length [m]	157.90 m
Boom length [m]	52.1 m	Weight of slings [t]	0 t	Underfloor height [m]	-112.8 m
Boom extension sequence [%]	100-100-100-100-100	Working radius [m]	25.0 m	Center of gravity X (to slewing center [m])	
Counterweight [t]	16.5 t	Boom angle [°]	64.0 °	Center of gravity Y (to slewing center [m])	
Outrigger base [m]	6.4 m	Lifting load [t]	1.29 t		

GROUND FORCES [t]

Slewing angle [°]	Outrigger			
	1	2	3	4
44.2	17	15	9	10
127.4	11	22	12	6
229.8	6	12	22	11
312.9	10	9	15	17

Maximum ground bearing load in service in working area

9. Load Details

9.1 LOAD DETAILS (as provided by Weatherproofing)

Load Description	Dimensions - Metres L-W-H	Weight KG	C of G	Sliding Arrangements
Roof Sheets	6m x 1.5m	200kg	Central	To be cradled with 6m 3te webbing slings.

9.1 Additional Lifts

During the works additional smaller lifts may be required.

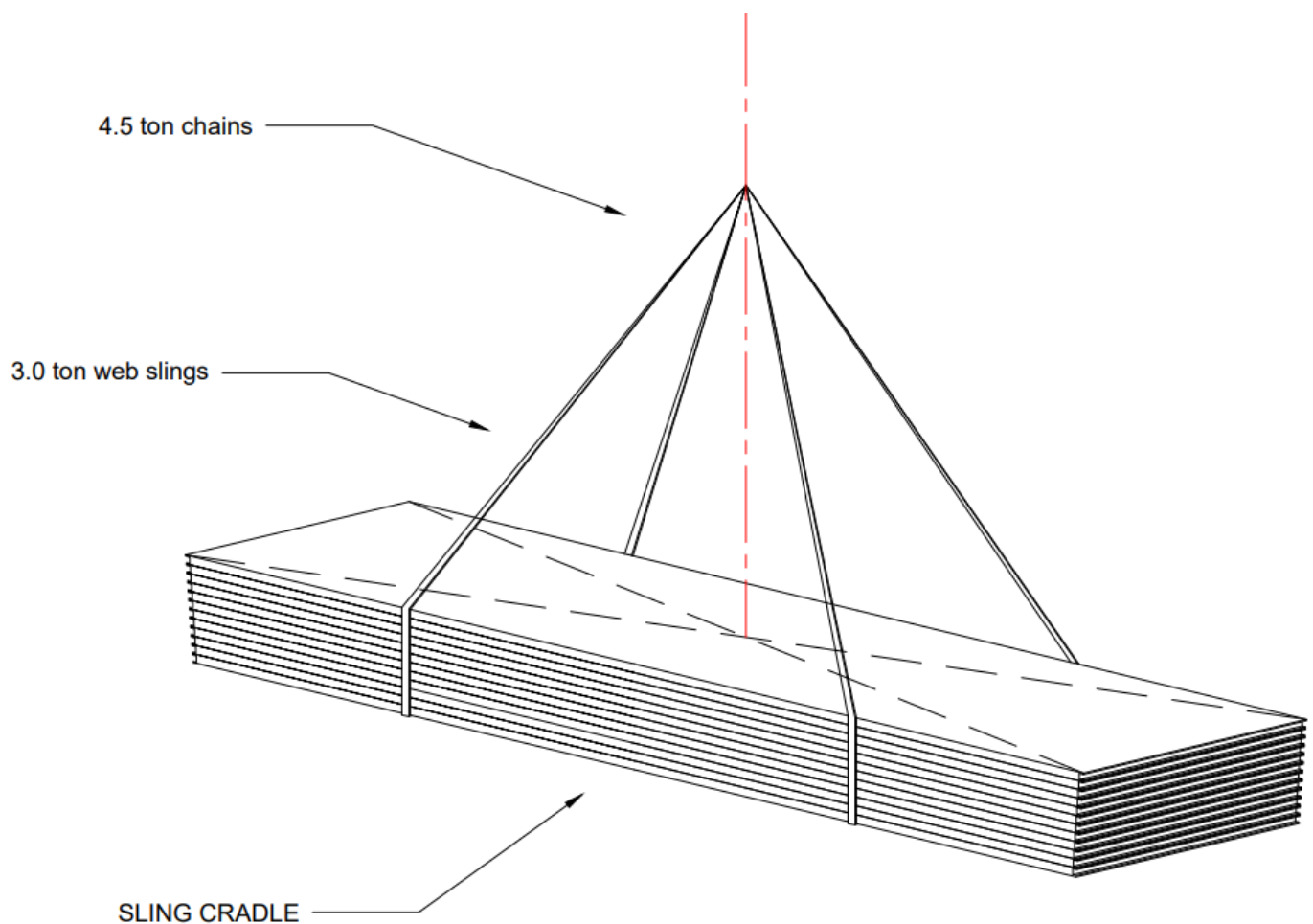
The Crane Supervisor will assess the requirements for any further items that may require lifting. These additional lifts will be determined as basic or standard lifts and a suitable procedure/lifting tackle arrangement will be agreed on site and recorded below in the box's provided. Should the Crane Supervisor determine them to be classed as complex lifts or the crane require moving to another location then he must liaise with the Appointed Person to agree a suitable procedure and lifting arrangement.

Load Description	Dimensions - Metres L-W-H	Weight KG	jib length	Radius	SWL	Slinging Arrangements

9.2 LIFTING TACKLE

All Lifting Tackle certification to be available on site. The Lift Supervisor shall inspect all Lifting Tackle prior to each use. Any Lifting Tackle supplied by Weatherproofing to be fit for purpose, fully certified and in good condition. Any damaged tackle shall be marked "Not for Use" and returned to the depot for disposal. Nylon slings must be protected from sharp edges using sling guards or suitable packing.

9.3 SLINGING METHODS



9.4 CALCULATED MAXIMUM LIFT WEIGHT

	<i>Main crane</i>	Choose an item.
Weight of load	200kg	
Weight of lifting tackle	100kg	
Weight of cranes hook block	190kg	
Gross lift weight	490kg	

9.5 CALCULATED MAXIMUM RADIUS

Stand off from nearest object	12m	
In - Reach	13m	
Lift Height	26m	
Maximum Radius	25m	

9.6 MAXIMUM CALCULATED LIFT

Gross lift weight	490kg	
Radius	25m	
Safe working load at radius	4200kg	
Safety Margin	3710kg	
Crane utilisation	11.67%	

9.7 LIFT AT MAXIMUM RADIUS

In the event that any loads (lighter than the maximum load) are required to be lifted to a greater radius then the maximum load will be as follows:

Gross lift weight	n/a	
Maximum Radius		
Safe working load at max radius		
Safety Margin		

10. Crane Lifting specification

Tadano ATF 70G-4L

City Lifting

CRANE HIRE SPECIALISTS

Capacity Charts

www.citylifting.co.uk

   16.5t DIN/ISO/EN													
m	11.1 m*	11.1 m	14.7 m	18.5 m	22.3 m	26.2 m	30.0 m	33.8 m	37.6 m	41.5 m	45.3 m	49.1 m	52.1 m
2.5	70.0**	57.7**											
3.0	59.3**	52.5**	50.4	49.0									
3.5	54.4**	48.2	46.2	44.8	39.9								
4.0	50.2	44.4	42.6	41.3	39.9	30.8							
4.5	44.2	41.1	39.6	38.3	37.4	28.7	23.9						
5.0	39.5	37.4	37.2	35.7	34.9	26.9	23.9	17.6					
6.0	32.3	31.3	31.5	31.2	30.6	26.2	22.8	18.6	14.2				
7.0	27.2	26.8	26.9	26.6	26.7	26.0	20.8	18.6	14.2	10.6			
8.0	23.4	23.1	23.3	23.0	23.1	23.5	20.4	17.6	14.2	10.6	8.3		
9.0			20.6	20.0	20.1	20.5	19.0	16.4	14.2	10.6	8.3	6.8	
10.0			18.2	17.6	18.4	18.1	17.8	15.1	13.3	10.6	8.3	6.8	5.8
12.0			14.4	14.8	14.7	14.7	14.8	12.9	11.7	10.6	8.3	6.8	5.8
14.0				11.4	11.7	11.8	11.5	11.1	10.4	9.5	8.3	6.8	5.8
16.0				9.0	9.6	9.5	9.1	8.9	9.1	8.5	7.9	6.8	5.8
18.0					7.9	7.7	7.6	7.8	7.4	7.4	7.1	6.6	5.8
20.0					6.4	6.4	6.8	6.5	6.2	6.1	6.3	6.0	5.7
22.0						5.9	5.7	5.4	5.6	5.5	5.2	5.3	5.3
24.0						3.8	4.9	4.9	4.8	4.7	4.7	4.4	4.5
26.0							4.2	4.4	4.1	4.0	4.0	3.7	3.8
28.0								3.8	3.8	3.6	3.4	3.2	3.2
30.0								3.3	3.3	3.1	2.9	2.7	2.7
32.0									2.8	2.7	2.5	2.2	2.3
34.0									2.5	2.3	2.2	1.9	1.9
36.0										2.0	1.8	1.6	1.6
38.0										1.7	1.5	1.3	1.3
40.0											1.3	1.0	1.0
42.0											1.1	0.8	0.8
44.0												0.6	0.6

11. Variation from Method Statement

Any and all significant variations to this Method Statement including drawings, must be notified to the Appointed Person before proceeding with the Lift. The details of the variations, together with the comments of the Appointed Person, must be noted below.

Variation Details	Time Contacted A. P	Initials
Comments from A. P	Permission to proceed	Initials
Variation Details	Time Contacted A. P	Initials
Comments from A. P	Permission to proceed	Initials
Variation Details	Time contacted A. P	Initials
Comments from A. P	Time contacted A. P	Initials

All other details are as per the Method Statement		
Crane supervisor	Signature	Date

If there are more than three variations the City Lifting Lift Supervisor is to discuss the situation with the Appointed Person.

All variations are to be of a lesser value than the main lift which City Lifting has been contracted to carry out and which has been specifically detailed within the main load calculation area of this method statement.

12. Placement of additional mats

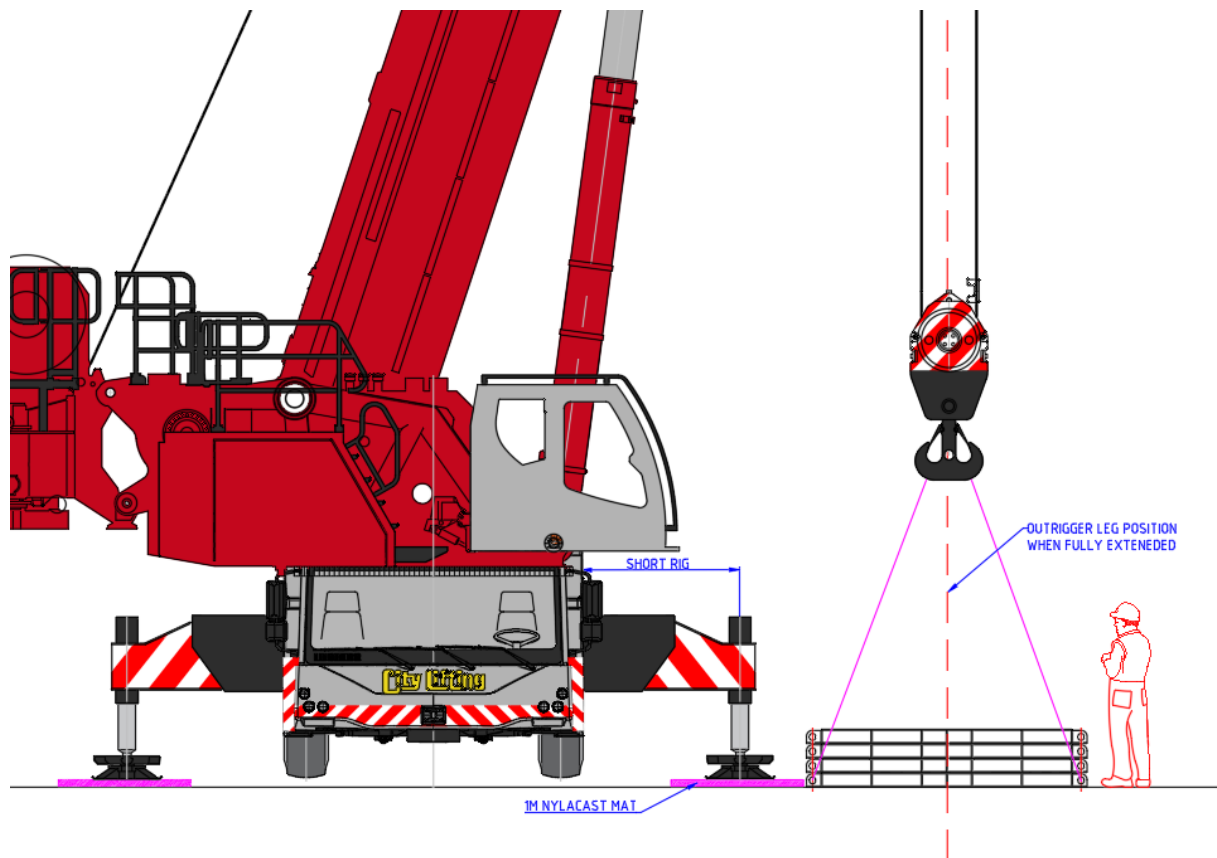
Any additional Mats for the rigging of the mobile crane are to be delivered by Hiab vehicle or Articulated lorry to site depending on size and weight. These extra mats are to be placed into position as per the attached lifting plan. All mats to be installed by the on-site lifting team at all times.

Final Matting position to be agreed by the lifting supervisor before the Mobile crane fully rigs on site.

12.1 Mobile crane Additional Mat installation

Mobile crane Operator to configure the crane on short rig duties on 1m-1m Nyla cast mats to allow the additional mats to be placed into the positions shown on the attached lifting plan. Any additional protection will be placed under the crane mats as required by using Etha foam sheets at 2.5m x 0.5m lengths.

Delivering Hiab vehicle is to reverse up to the Mobile crane to within its safe working load for removal of the mats. Mats to be removed using the mobile crane's lifting tackle only.



12.2 Hiab Additional Mat installation

Lorry Driver – Operator, to configure the Hiab crane on full rig duties on 1m-1m Nyla cast mats to allow the additional mats to be placed into the positions shown on the attached lifting plan. Any additional protection will be placed under the crane mats as required by using Etha foam sheets at 2.5m x 0.5m lengths. Hiab vehicle to be operated by City Lifting delivery driver on a CSCS or ALME qualification. All lorries to be accessed via fixed steps and handrail system when working on the back of transport bed.

13. Crane set up procedure

- 13.1** The crane together with the associated operating crew will arrive on site at the specified and agreed date and time.
- 13.2** Lift Supervisor will meet client representative and carry out a toolbox talk with all personnel involved in the lifting operation. Once this has been completed, all involved in the lifting operation must sign the register in this method statement. Any site inductions that are required will also be carried out.
- 13.3** Any traffic management, ancillary pans, mats or temporary roadways etc., that are required to be positioned by Hiab or other means, prior to the crane manoeuvring into position, will be done under the instruction and guidance of the Lift Supervisor.
- 13.4** The Crane will then manoeuvre into position as shown on attached drawing. All crane or manoeuvring to be carried out under the guidance of a member of the lifting crew.
- 13.5** The Crane will then commence set-up procedure and rig to the configuration stated in this method statement. All personnel assisting with rigging will be under the guidance of the Crane Operator or operators, at all times. The set-up procedure will be carried out as per crane manufacturer's manual.
- 13.6** Upon completion of the rigging procedure the operator will install the anemometer & carry out procedural safety and computer checks, implementing any specific operational requirements, i.e. slewing restrictions, short rig configurations, or fly jib configurations etc.
- 13.7** The Lift Supervisor will then carry out a radio communications check between the Crane Operator and the lifting crew making sure the radios are all synced to the same channel and working correctly.
- 13.8** The hook block of the crane will then be lowered and required lifting tackle placed on the hook. The Lift Supervisor will ensure that the correct lifting tackle has been selected.
- 13.9** The Crane Operator will then be instructed to hoist lifting tackle clear of any obstructions and position the crane ready for the first required lift. Any delivery / removal vehicles will now be positioned adjacent to the crane as per attached drawings, ready for work to commence.

14. Lifting Procedure

- 14.1** All slinging will be carried out by fully trained, certificated and experienced slingers. Slinging methods adopted will be as specified by load manufacturers, using specified lifting points or, in the event that no manufacturer instructions or lifting points are provided, our slingers will work to the Appointed Persons recommendations and to Health & Safety executives guidelines for safe slinging of lifting loads, The lifting tackle used will be as stated in this method statement and will have current test certification available for inspection.
- 14.2** Slinger / Signaller will instruct the Crane Operator to lower the hooks with lifting tackle to allow Slinger / Signaller to attach lifting tackle to load.
- 14.3** Slinger / Signaller will issue instructions via hand signals or two-way radios to the Crane Operator to position crane over first required lift. Slinger / Signaller will plumb boom head or heads directly over the load.

- 14.4** Upon completion of the attachment of the lifting tackle to the load, the Slinger / Signaller will instruct the operator by hand signals or radios, to hoist up to take the slack out of the lifting tackle. Once the lifting tackle has started to take the weight and has tightened the Slinger / Signaller will instruct the operator to stop hoisting. The Slinger / Signaller will then carry out a check around the load to ensure that the lifting tackle is sitting correctly and angled correctly and also check any lifting points for security and correct operation.
- 14.5** Once lifting tackle check has been carried out the Slinger / Signaller will instruct the Crane Operator to hoist the load to a height that allows the lifting tackle to take the full weight of the load and again instruct the Operator to stop hoisting when load is just airborne to allow a final check of the lifting tackle and lifting points and also check the structural integrity of the load.
- 14.6** Once the Slinger / Signaller is confident that the lifting tackle and the load are secure, any tag lines that may be required will be attached to the load. Instructions will then be given to the Crane Operator to hoist the load clear of any obstructions, ensuring that any site-specific risks are monitored throughout the hoisting procedure.
- 14.7** Once the load is clear of all obstructions, the Slinger / Signaller will give instructions to the Crane Operator to manoeuvre the load as required to the required landing area.
- 14.8** In the event that the landing area is at a different level or should access not permit, the Slinger / Signaller will pass control of the crane signalling to a second strategically placed Slinger / Signaller at the landing area.
- 14.9** The landing area Slinger / Signaller will then commence giving instructions to the Crane Operator to manoeuvre and lower the load to the required landing point, ensuring that any site-specific risks are monitored throughout the landing procedure.
- 14.10** The Slinger / Signaller will guide the load to the correct orientation as the load becomes contactable by hand or by any attached tag lines.
- 14.11** The Slinger / Signaller will instruct the Crane Operator or operators to lower the load to just above the landing position and then instruct the operator or operators to stop lowering to allow a Weatherproofing representative to confirm position and orientation of load prior to landing.
- 14.12** Once the position is confirmed by Weatherproofing, the Slinger / Signaller will instruct the operator to slowly lower load into position. Once landed, a final position check will be made before the Slinger / Signaller instructs the operator to lower off all load weight.
- 14.13** Slinger / Signaller will instruct the Operator to lower off, to allow the lifting tackle to be released from the load. In the event that any load is too large or high for safe access by ladder, powered access equipment or a lifted man basket will be used to remove slings from loads.
- 14.14** Once lifting tackle is detached from the load the Slinger / Signaller will give instructions to the Crane Operator to slowly hoist the lifting tackle clear of the load and any site-specific risks ensuring that no snagging occurs.
- 14.15** Once the lifting tackle is clear of all obstructions the Slinger / Signaller will instruct the Crane Operator to manoeuvre the crane or cranes ready for any further lifts that may be required and hand over to another slinger / signaller where required.

15. De-Rig Procedure

- 15.1** Upon completion of the lifting operation, the Lift Supervisor will instruct the Crane Operator to load any ancillary equipment or heavy lifting tackle onto relevant transport ready for removal from site.
- 15.2** The Lift Supervisor will then instruct the Crane Operator or operators to lower all other lifting tackle to ground to allow removal from hook.
- 15.3** The Crane Operator will then instigate the removal of any lifting tackle stored on the crane and then commence the de-rig procedure.
- 15.4** All personnel assisting with de-rigging will be under the guidance of the Crane Operator at all times. The de-rig procedure will be carried out as per crane manufacturer's manual.
- 15.5** Once crane has de-rigged and all safety checks for road travel have been carried out, the Crane Operator will manoeuvre the crane off site, under the guidance of a member of the lifting crew. The crane will then proceed on its way.
- 15.6** The remaining members of the Lifting Crew will ensure that any ancillary mats or equipment etc. are removed from site or placed in an appropriate position ready for collection prior to them leaving site.
- 15.7** Any traffic or pedestrian management arranged by City Lifting will be unimplemented and removed. The Lift Supervisor will inform Weatherproofing, Weatherproofing representative, or site representative of the completion of the operation and address any issues that may be raised and inform of any equipment left on site for later collection.

16. Emergency Procedure

- 16.1** In the event of any accident or emergency on site, City Lifting staff will follow the Below procedures before starting the lifting procedure again:
- Lifting Supervisor to report all incidents or accidents to Site Management.
 - Lifting Supervisor to stop all lifting operations and make the crane and lifting area safe before reporting.
 - Hire Desk to be contacted on 01708 805550 (out of hours main phone line diverts to Company Director)
 - In case of a major incident, local authorities will be notified as well as site management and City Lifting M.D
 - All incidents minor or major must be recorded in the on-site accident book.

Once the incident / accident has been reported all lifting operations will cease until the Site Management and Lift Supervisor have agreed that the lift is safe to be completed.

16.2 EMERGENCY CONTACT DETAILS

Steve Goodey	Appointed Person	07484 061183
Bob Jones	Director	07976 927091
Trevor Jepson	Managing Director	07973 612860

17. Risk Assessment Key

Client	Weatherproofing	Date	21/07/2023	Rev Date	Click or tap to enter a date.
Site Ref	SG2307213	Assessor	Steve Goodey	Signed	<i>SRGoodey</i>

RISK ASSESSMENT - MANAGEMENT OF HEALTH AND SAFETY AT WORK REGULATION 1999

The aim of this risk assessment is to highlight any possible hazards that are likely to occur during the lifting operation. Upon its completion with either a “High, Medium, Moderate or Low in the Risk Rating box and either a Moderate or Low in the Risk Managed box, the document converts to a site-specific assessment

APPROPRIATE PPE TO BE WORN AT ALL TIMES.

Risk Matrix High - Medium - Moderate - Low (Risk)						
Severity X Likelihood = Risk Rating		Likelihood				
		Certain (5)	Very Likely (4)	Likely (3)	May Happen (2)	Unlikely (1)
Severity	Death (5)	25	20	15	10	5
	Major Injury (4)	20	16	12	8	4
	Over 7-day Injury (3)	15	12	9	6	3
	Minor Injury (2)	10	8	6	4	2
	No Injury (1)	5	4	3	2	1

(1 – 4) Low Risk	Manage by routine procedures
(5 – 9) Moderate Risk	Specify Management Responsibility – Review control measures
(10 – 16) Medium Risk	Further Controls will be required – Improve control measures
(17 – 25) High Risk	Discontinue Operation – Immediate Corrective action required

No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R
17.1	<u>Access to and Egress from site.</u>	Crushing, serious injury, fatal injury. Damage to plant, equipment and property. Plant and vehicle coming into contact with each other or people and property.	City Lifting crane operators, slinger/signaller Crane supervisor and all other City Lifting personnel. Any Other personnel within access and egress route.	5	4	20	Access to and egress from all site compounds shall under space restriction areas, be controlled by a Banksman. At all times pedestrians and vehicles must be segregated with particular attention being paid to restricted and confined space areas. When necessary the correct high visibility vests shall be worn. Each depot compound must have a yard plan available and displayed giving information on arrangements. Any approach to vehicles must be from the front so that the drivers can be aware of your presence. Flashing amber lights and audible alarms when reversing.	4	2	8
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R
17.2	<u>Ground Conditions.</u> Poor Ground Conditions Uneven Ground Interface with others	Transport Traction Unsuitable Equipment Damage to plant or property	Crane Operator & Persons within the radius of the crane	3	4	12	Access to and egress from all site compounds shall under space restriction areas, be controlled by a Banksman. At all times pedestrians and vehicles must be segregated with particular attention being paid to restricted and confined space areas. When necessary the correct and appropriate high visibility vests shall be worn. Ensure ground conditions are capable of withstanding pressures exerted. Area to be checked for overhead obstructions and underground hazards.	4	2	8
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R

17.3	Underground Services Electrical Cables Water Services Telecom Cables C.A.T.V. Services	Fire Explosion Flooding Asphyxiation Electrocution Disruption of services to public.	Crane Operator & Persons within the radius of the crane	5	3	15	In the event that digging may be necessary the area of works shall be checked for services through surveys, maps drawings, local authorities, scanning equipment, customer and clients. When necessary a permit to work system shall be implemented between the customer, client and management of persons carrying out the works. Where heavy vehicles have to cross over known areas of water pipes and gas pipes, steel plates shall be used to protect against crushing and fracturing. All supplies that may be at risk through crossing of heavy plant or digging shall be clearly marked up.	4	1	4
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R
17.4	Movement of Plant	Plant and vehicles coming into contact with each other, people and property. Collision, crushing or fatal injury.	Crane operators, slinger/signaller Appointed person Crane supervisor, All other City Lifting personnel and other personnel within movement area.	5	4	20	All site areas shall be controlled by use and display of signs, markings and speed limits, which shall be adhered to at all times. Movement of any plant in a restricted area shall be controlled by a Banksman. All plant shall be equipped with reversing alarms. All plant shall be parked in an orderly manner. All movement around Site shall be by designated routes. High visibility clothing to be worn by all. Sufficient lighting to be provided at all times.	4	2	8
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R

17.5	<u>Crane Positioning</u> Set up Procedure De-rig Procedure	Overturning of the crane Collapse of the ground beneath outriggers Injury to operator in control cab Injury to personnel in crane radius Possible damage to crane or property during set up or de-rig.	Crane operators, slinger/signaller Appointed person Crane supervisor, All other City Lifting personnel and other personnel within movement area.	5	3	15	Crane outriggers shall be deployed in accordance with manufacturers instruction. Outriggers shall only be deployed on firm level ground Outriggers feet shall be fully supported on purpose made pans or adequate timbers. When riggers are fully deployed the crane shall be level with all wheels clear on the ground. Sufficient ballast/counterweight shall be in place. R.C.I. shall be in good working control. Cab computer read out shall match that of the crane configuration. Cranes shall stop operating in wind speeds greater than those laid down by the manufacturer.	4	2	8
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R
17.6	<u>Loading / Unloading</u> City lifting transport or cranes Loading / Unloading Sites transport	Serious or fatal injury from trapping crushing and from falling objects. City Lifting personnel falling from ballast support vehicle when carrying out the positioning of crane counterweights & ancillary equipment on the deck of trailer	Crane operators, slinger/signaller Appointed person Crane supervisor, All other City Lifting personnel and other personnel within movement area.	5	2	10	Only the minimum number of persons to be engaged during the operation. Ensure no public interface during operations - make use of barriers, tape etc. Vehicles must be in good efficient state and repair. Loading/unloading to be controlled by Appointed Person or Crane Supervisor. Vehicles shall not be overloaded, only carrying the capacity that it was designed for. No vehicle shall be loaded to such an extent that it may interfere with the safe driving or operation of the equipment. When access and egress takes place on all trailers City lifting will make sure it is done by means of a ladder at all times.	4	2	8
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R

17.7	<u>Manual Handling Operations</u>	Injury to hands feet and back through cuts, bruising, sprains and strains	Crane operators, slinger/signaller Appointed person Crane supervisor, All other City Lifting personnel and other personnel within movement area.	2	3	6	Where possible the requirement to manual handle shall be eliminated by making use of mechanical means such as cranes, hoists, forklifts, sack barrows and other proprietary lifting, pushing and pulling tools. Where lifting, pushing and pulling cannot be avoided assessment shall identify a safe system of work which shall include the use of more than one person to carry out the required operation. Suitable periods of work and rest in order to prevent fatigue shall be planned into all operations. Use shall be made of Kinetic Lifting Techniques. Appropriate personal protective equipment shall be worn at all times.	2	1	2
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R
17.8	<u>Working at Height</u> Ladders Scaffolding Access Platforms Man Baskets	Personnel falling from height during crane assembly and or lifting operations. Personnel falling from delivery vehicles during slinging of loads. Operation of mechanical platforms. Personnel working from man baskets.	Crane operators, slinger/signaller Appointed person Crane supervisor, All other City Lifting personnel and other personnel within movement area.	5	4	20	Each crane shall carry a ladder of suitable length and strength to enable works at height to be carried out without the requirement to climb. Where not practical with ladders, operators shall wear a body harness's and shall be clipped on to the inertia reel which will be suspended from the mobile crane hook-block. Should neither of the above be practical, access shall be via access platform or in extreme cases a scaffold tower. The two latter are subject to provision of adequate working area. Work platforms or lifted man basket to be used where a height risk is deemed excessive. Access operators to be IPAF trained. Site to ensure all working areas at height are safe with suitable guard rails.	4	2	8
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R

17.9	Protection of the public & other trades	Public entering into the works area. Public suffering injury due to entering the work area. Falling objects / lifting apparatus failure	Members of the public and other Trades personnel within the crane operating area.	5	3	15	The areas in which the lifting operation are to take place shall be completely barriered off by the client to prevent the public gaining access. Warning signs shall be displayed by our client at prominent places warning the public and other trades of danger from lifting operations. The lifting operations shall be halted immediately if any member of the public gains access inside the barriered area. Only members of the lifting operations team shall be permitted within the barriered area. Particular attention to be paid to the presence of site visitors, children & others viewing the lift in a public space. Do not lift over open footpaths or roads on site. Equipment & slings to have a valid thorough examination certificates in place, equipment & slings to be checked prior to use, competent plant operator and slingers only to carry out lifting operations/slinging Exclusion zones in place around lifting operations, no one to pass beneath suspended loads.	4	2	8
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R
17.10	Adverse weather Heat exhaustion other climate controls	Slipping on wet Surfaces Getting cold Fainting Passing out Skin burn Impaired vision	Crane operator & rigging crew. All other site personnel working on site.	3	2	6	Crane to be operated in accordance with BS7121 parts 1 & 3 with regard to weather conditions. In the event of lightning/thunder risk, all work will cease & the crane jib retracted & lowered as low as possible to reduce the chance of lightning strike. If possible, all work to cease in the event of a prolonged downpour. All personnel to take cover. Personnel to wear foul weather gear. In the event of high winds all work will cease when the maximum wind speed (as per these RAMS) is exceeded. All personnel to take cover & remove themselves from any high open areas and be aware of flying debris. Avoid prolonged periods in direct sunlight. Places of shade to be made available by the main contractor. Keep skin covered and apply high factor sun protection creams (30+).	2	2	4
No.	Identified Hazard	Risk	Persons at risk	S	L	R	Control Measure	S	L	R

17.11	<u>Slewing Restrictions</u> Low Headroom Slew Restrictions Neighbouring Rail Neighbouring Power Lines	Damage to crane or property if crane boom or superstructure comes into contact with any restriction. Infringement of unauthorised air space. Working in restricted workspace. Collision with Buildings	Crane operator & rigging crew. All other site Personnel working on site. Public	5	4	20	Full assessment of any risks made and addressed at site inspection. Safe system of work adhered to where implemented. Work to zone alarms where implemented with slew restrictions in place Implement crane swl computer working restrictions. All relevant authorities contacted and advised, and any restrictions adhered to. Any electrical risks to be assessed and addressed at site inspection. Client to ensure City Lifting are made aware of any risks. Safe system of work implemented and adhered to. All operational personnel to be made aware of risk. Statutory regulations to be followed unless otherwise authorised. Implement crane swl computer working restrictions. Crane position set to limit collapse radius from rail boundary Crane boom length set to limit collapse radius from rail boundary Crane boom length set to ensure crane/load cannot impact rail assets or trains (m radius limit) Limited working radius ensures works are clear of 2.75m from live OLE Crane & GBP down rated for works adjacent to NR infrastructure (crane utilisation =) (ground loadings = of acceptable loadings for roadways)	3	2	6

18. Toolbox Talk Attendance

Date Held:		Time Held:	
- Lifting Procedure - PPE Requirements - Crane Set Up - Method of Lifting		- Lifting Tackle - Site Specific Hazards - Emergency Procedure	
Name:	Trade:	Company:	Signature:
	Crane supervisor	City lifting	
	Crane operator	City lifting	
	Slinger signaller	City lifting	
Client's Acceptance of this method statement			
Client's Signature		Date	
Supervisor's Signature		Date	

Pre Lift-Check Points Tick boxes to confirm the following (where applicable) have been checked			
	y	n	
Crane(s) test certificates	☐	☐	Site induction
Crane thorough examination report	☐	☐	Method Statement & Drawings Issue
Operator weekly inspection form	☐	☐	Risk Assessment issued
Manufacturer instruction manual	☐	☐	Slinger's Competency certification
Test certificates / thorough exam Reports for all lifting accessories	☐	☐	Operator(s) CPCS approved and Crane specific certificates
Spill pack available	☐	☐	Communication Method identified
Toolbox talk delivered and recorded	☐	☐	Appropriate PPE

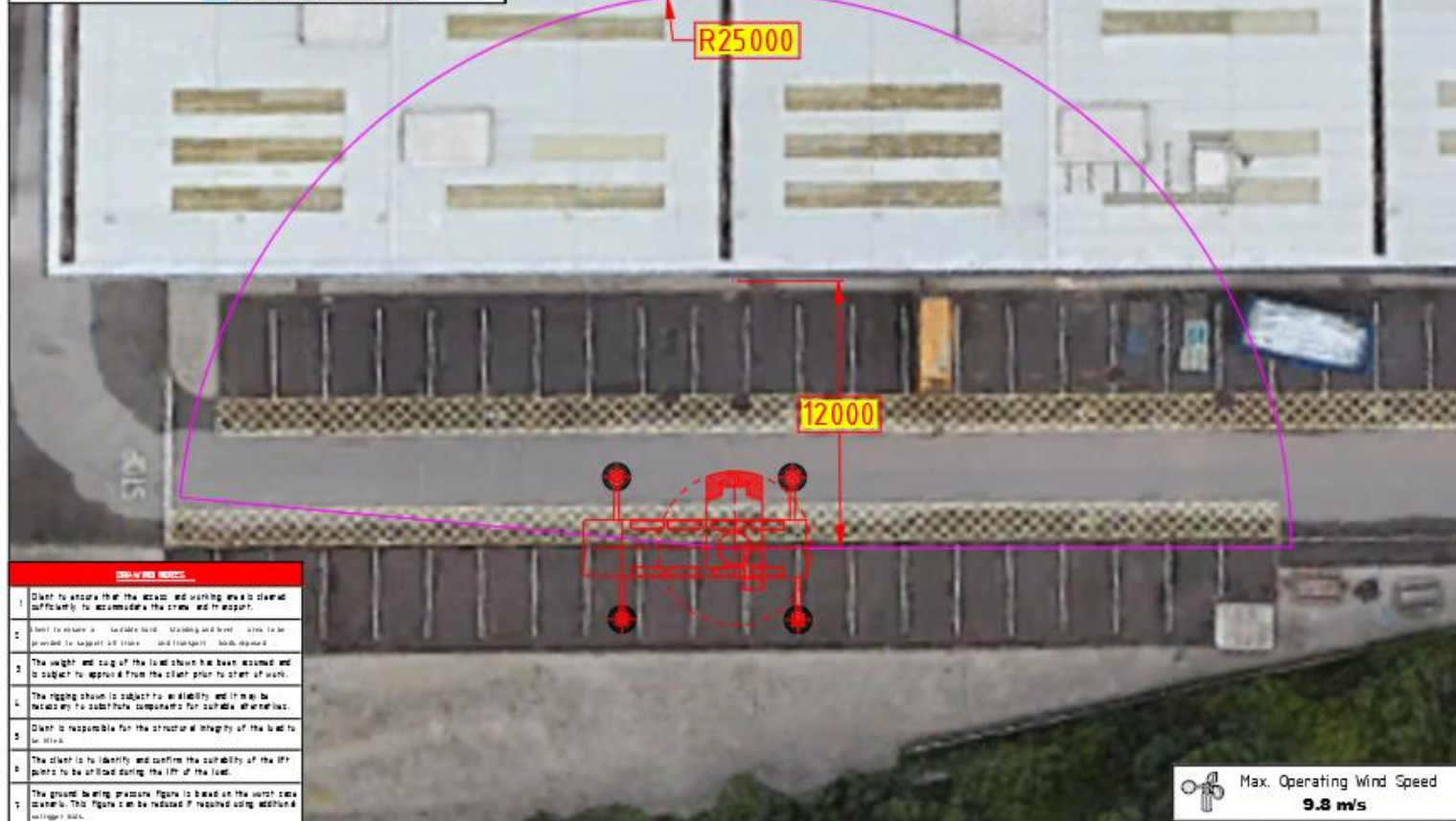
Erection Checklist Tick boxes to confirm (where applicable) completion			
	y	n	
Working area cordoned off	☐	☐	Lift performed as Method Statement
Cranes set in correct location	☐	☐	Item accepted by client
Crane limits & load indicator OK	☐	☐	Rigging released as Method Statement
Rigging fitted to item as detailed	☐	☐	Cranes de-rigged
Weather within acceptable limits	☐	☐	Site cleared

Job Completion Sign off

I confirm that the lifting operations have been performed as detailed in this Method Statement and Risk Assessment.					
Full Name		Signed		Date	
Comments					

19. Lifting Plans and Supporting Documentation

TADANO City Lifting			
OUTRIGGER REACTION			
ATF 70G-4 EU Stage IV (52.1m) (Standard)			
Boom without Jib			
CONFIGURATION			
30 - Configuration (m)	g	Maximum ground bearing load in service in working state	1
30 - Adjustment angle (°)	α	Truckbed tilt	2
Boom length (m)	52.1 m	Working radius (m)	25.0 m
Boom extension (m)	120-100-100-100	Boom angle (°)	94.0°
Counterweight (t)	16.5 t	Upright level (t)	1.29 t
Outrigger level (m)	0.4 m		
GROUND PRESSURE (t)			
Swinging angle (°)	1	2	3
0.0	19	15	10
10.0	15	10	8
20.0	9	10	11
30.0	10	9	15



MAIN CRANE DATA (M3 ISO 754)	
Crane Model	Tadano M3 ISO 754
Lifting Configuration	10
Outrigger Base Center	10 x 10 m
Counterweight	16.5 t
Boom Base Length (m)	52.1
Telescopic Sections	120-100-100-100
Hookblock Type	ATM-1
Outrigger Pads	10
Additional Data	10

LIFT DETAILS - CRITICAL LIFT	
Item To Be Lifted	Red Steel
Radius	25.0 m
Item Weight	1.2 t
Lifting Time	0.5
Hookblock	ATM-1
Total Lift Weight	1.2 t
Crane Capacity	1.2 t
70 SF Capacity	1.2 t

OUTRIGGER LOAD / MAT INFORMATION	
Item Outrigger Force	1.2 t
Crane Model	100 x 100 x 100
Outrigger Pads	10 x 10 m
70 SF Capacity	1.2 t
70 SF Capacity	1.2 t

Original Date: 01/10/2020	
Client:	Weatherproofing
Site:	100 x 100 x 100 m
Project:	Lifting materials to roof level
Model:	ATM-1
Status:	ISSUE
Scale:	PLAN VIEW
Job No:	100 x 100 x 100
Sheet No:	1
Rev:	0
City Lifting	
100 x 100 x 100 m	